Lab 01

Week 9

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1 Oracles, the Deutsch Algorithm and the Deutsch-Jozsa Algorithm

The whole lab is done on local jupyter notebook. All codes and experiments are performed in the notebook provided in the appendix.

The answers and code solutions are provided
in the appendix located on the next page…

APPENDIX 1

Lab 01 | Code Solution

lab-01-code

October 18, 2026

1 Lab 1: Oracles, the Deutsch Algorithm and the Deutsch-Jozsa Algorithm

Adapted from the IBM Quantum lab "Oracles and the Deutsch-Jozsa Algorithm".

Duration: 90 minutes. The theory was covered in the lecture; in this lab you build and run the circuits.

Part
Installation
Part 1) Classical oracles
Part 2) The Deutsch algorithm (1 input bit)
Part 3) The Deutsch-Jozsa problem
Part 4) The Deutsch-Jozsa algorithm (n input bits)
Problems 1–10

How to use this notebook - **Exercises** are highlighted in green boxes. Write your code between the `### FILL YOUR CODE IN HERE` markers in the cell below each box. - **Check** cells test your code. If they print a success message you can move on; if they raise an error, read the message. - **Q** — answer the question in the cell below it. - **Optional** — do these only if you are ahead of the schedule.

1.1 Installation

Run this cell first to check your Python version.

[1]:
```
import sys

recommended = (3, 10)
current = sys.version_info[:2]

if current < recommended:
    print(f"You are using Python {current[0]}.{current[1]}. Python␣
 ↪{recommended[0]}.{recommended[1]} or later is recommended.")
else:
    print(f"Python {current[0]}.{current[1]} - looks good!")
```

```
Python 3.10 - looks good!
```

Next, run the cell below to install and upgrade the required packages. If the upgrade process causes dependency conflicts with packages already installed in your local environment, **create a new Python environment** (Python 3.12 recommended) and continue the lab there.

For guidance on setting up a clean environment and installing Qiskit, refer to the IBM Quantum documentation: https://quantum.cloud.ibm.com/docs/guides/install-qiskit

[2]:
```
# Install required packages.
# Run this cell the first time you use this notebook (or in a new environment),
# then comment the line below out on subsequent runs.

# %pip install --upgrade 'qiskit[visualization]>=2.5.0' qiskit-aer matplotlib
```

Once you have installed and upgraded the libraries, please restart the Python kernel and rerun the notebook starting from the Import Libraries cell below.

[3]:
```
# Import Libraries
import random
import numpy as np
from IPython.display import display

# Importing standard Qiskit libraries
import qiskit
import qiskit_aer
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator
from qiskit.quantum_info import Statevector, Operator
from qiskit.visualization import plot_bloch_multivector, plot_histogram
from matplotlib import pyplot as plt

sim = AerSimulator()

def run(qc, shots=1024):
    """Simulate a circuit and return the measurement counts."""
    return sim.run(transpile(qc, sim), shots=shots).result().get_counts()

print(f"Qiskit version: {qiskit.__version__}")
print(f"Qiskit Aer version: {qiskit_aer.__version__}")
```

```
Qiskit version: 2.5.2
Qiskit Aer version: 0.17.2
```

Run the next cell. It defines the check functions — you do not need to read it.

[4]:
```
# Check functions (you do not need to read this cell)

def _state(qc):
    return Statevector(qc.remove_final_measurements(inplace=False))
```

```
def _counts(qc, what):
    assert qc.count_ops().get('measure', 0) > 0, f"{what} has no measurement -␣
 ↪did you forget qc.measure(...)?"
    return run(qc)

def check_classical_solver(solver):
    worst = {}
    for n in range(1, 7):
        for seed in range(300):
            f = ClassicalOracle(n, seed)
            truth = "constant" if len(set(f._table)) == 1 else "balanced"
            answer = solver(f)
            assert answer == truth, f"n = {n}, seed = {seed}: you answered␣
 ↪{answer!r}, but the function is {truth!r}."
            assert f.queries <= 2**(n - 1) + 1, (f"n = {n}, seed = {seed}: you␣
 ↪used {f.queries} queries. "
                                                 "You can always decide with␣
 ↪fewer - when can you stop?")
            worst[n] = max(worst.get(n, 0), f.queries)
    print("classical_solver is always correct for n = 1 ... 6")
    print("   most queries you needed:", ", ".join(f"n={n}: {q}" for n, q in␣
 ↪worst.items()))

def check_deutsch_oracle(make_oracle):
    expected = {0: [0, 0], 1: [1, 1], 2: [0, 1], 3: [1, 0]}
    for case, table in expected.items():
        U = Operator(make_oracle(case)).data
        for x in (0, 1):
            for y in (0, 1):
                out = ((y ^ table[x]) << 1) | x      # basis index of |q1 q0> =␣
 ↪|y XOR f(x), x>
                assert np.isclose(abs(U[out, (y << 1) | x]), 1), (
                    f"case {case}: U_f |x={x}>|y={y}> should give |x={x}>|y={y␣
 ↪^ table[x]}>")
    print("deutsch_oracle is correct for all four functions")

def check_deutsch_circuit(make_circuit):
    for case in range(4):
        counts = _counts(make_circuit(deutsch_oracle(case)), "deutsch_circuit")
        expected = '0' if case < 2 else '1'
        kind = 'constant' if case < 2 else 'balanced'
        assert counts == {expected: 1024}, f"f{case} is {kind}: expected␣
 ↪{{'{expected}': 1024}}, got {counts}"
        print(f"   f{case} ({kind}): {counts}")
    print("deutsch_circuit is correct: one query tells constant (0) from␣
 ↪balanced (1)")
```

```
def check_psi_0(psi_0, n):
    assert _state(psi_0(n)).equiv(Statevector.from_label('-' + '0' * n)), \
        f"psi_0 should be |-> on q_{n} and |0> on q_0 ... q_{n-1}"
    print("psi_0 is correct")

def check_psi_1(psi_1, n):
    assert _state(psi_1(n)).equiv(Statevector.from_label('-' + '+' * n)), \
        f"psi_1 should be |-> on q_{n} and |+> on q_0 ... q_{n-1}"
    print("psi_1 is correct")

def check_psi_2(psi_2, oracle, n):
    expected = Statevector.from_label('-' + '+' * n).evolve(Operator(oracle))
    assert _state(psi_2(oracle, n)).equiv(expected), "psi_2 should be the␣
 ↪oracle applied once to psi_1"
    print("psi_2 is correct")

def check_full_circuit(full_circuit, n):
    const = QuantumCircuit(n + 1, name="constant"); const.x(n)
    c = _counts(full_circuit(const.to_gate(), n), "full_circuit")
    assert c == {'0' * n: 1024}, f"for a constant oracle you should always␣
 ↪measure {'0' * n}, got {c}"
    for i in range(n):                          # balanced f(x) = x_i: one test␣
 ↪per input qubit
        bal = QuantumCircuit(n + 1, name=f"x{i}"); bal.cx(i, n)
        b = _counts(full_circuit(bal.to_gate(), n), "full_circuit")
        assert '0' * n not in b, (f"for the balanced oracle f(x) = x_{i} you␣
 ↪measured {'0' * n}, so it looks "
                                  f"constant - are a Hadamard and a measurement␣
 ↪applied to q_{i}? got {b}")
    print("full_circuit is correct")
```

1.2 Oracles and query complexity

Many quantum algorithms revolve around the notion of so-called *oracles*. An oracle is a function that we can only use as a "black box": we can ask it questions (*queries*) but cannot look inside. We want to find out a property of this function. The *query complexity* is the minimum number of queries needed to determine that property.

In this lab the hidden function f is guaranteed to be one of two kinds, and we must decide which: - **constant** — returns the same value (all 0 or all 1) for every input - **balanced** — returns 0 for exactly half of the inputs and 1 for the other half

We start with the simplest case, a function of one bit (the **Deutsch problem**), and then generalise to n bits (the **Deutsch-Jozsa problem**). Each time we compare how many queries a classical and a quantum computer need.

1.3 Part 1) Classical oracles

1.3.1 1.1 All functions of one bit

A function $f : \{0, 1\} \to \{0, 1\}$ has only four possibilities. Run the cell to print their truth tables.

[5]:
```
one_bit_functions = {
    "f0": lambda x: 0,
    "f1": lambda x: 1,
    "f2": lambda x: x,
    "f3": lambda x: 1 - x,
}

print("f   | f(0) f(1)")
for name, f in one_bit_functions.items():
    print(f"{name}  |  {f(0)}    {f(1)}")
```

```
f   | f(0) f(1)
f0  |  0    0
f1  |  1    1
f2  |  0    1
f3  |  1    0
```

Q1. Which of f_0, f_1, f_2, f_3 are constant and which are balanced? You query the black box once and get $f(0) = 0$. Which functions are still possible — can you already tell whether f is constant or balanced? What if you got $f(0) = 1$?

Your answer:

- **Constant:** f_0 ($f(0) = f(1) = 0$) and f_1 ($f(0) = f(1) = 1$).
- **Balanced:** f_2 ($f(0) = 0, f(1) = 1$) and f_3 ($f(0) = 1, f(1) = 0$).

If we query once and get $f(0) = 0$, the function could still be f_0 (constant) or f_2 (balanced), so we **cannot** tell yet. If we get $f(0) = 1$, it could still be f_1 (constant) or f_3 (balanced), so again we cannot tell. Classically, one query is never enough: we always need a second query, $f(1)$.

1.3.2 1.2 A classical black box

`ClassicalOracle(n)` hides a random function of n bits that is guaranteed to be constant or balanced. You can only call it as `f(x)`, where `x` is an integer from `0` to `2**n - 1`. It counts every call in `f.queries`. (Please don't peek at `f._table` — that would be opening the box!)

[6]:
```
class ClassicalOracle:
    """Black box: you can call f(x) but not look inside. Counts every query."""
    def __init__(self, n, seed=None):
        rng = random.Random(seed)
        self.n, N = n, 2**n
        if rng.random() < 0.5:                          # constant
            self._table = [rng.randint(0, 1)] * N
        else:                                           # balanced
            ones = set(rng.sample(range(N), N // 2))
```

```
            self._table = [int(x in ones) for x in range(N)]
        self.queries = 0

    def __call__(self, x):
        self.queries += 1
        return self._table[x]
```

Exercise 1: Write a classical solver

Write classical_solver(f) that returns 'constant' or 'balanced'. It must always be correct and use as few queries as possible.

Hint: when can you stop early? How many equal answers in a row guarantee "constant"?

[7]:
```
def classical_solver(f):
    """Return 'constant' or 'balanced'. f.n is the number of input bits; inputs␣
 ↪are 0 ... 2**f.n - 1."""
    ### FILL YOUR CODE IN HERE
    first = f(0)
    # a balanced function has only 2**(n-1) equal outputs, so 2**(n-1) + 1␣
 ↪equal answers prove "constant"
    for x in range(1, 2**(f.n - 1) + 1):
        if f(x) != first:
            return 'balanced'
    return 'constant'
    ###
```

[8]:
```
# Check
check_classical_solver(classical_solver)
```

```
classical_solver is always correct for n = 1 … 6
   most queries you needed: n=1: 2, n=2: 3, n=3: 5, n=4: 9, n=5: 17, n=6: 33
```

1.3.3 1.3 How many queries for n bits?

Run your solver on 2000 random oracles with $n = 6$ and look at how many queries it needed.

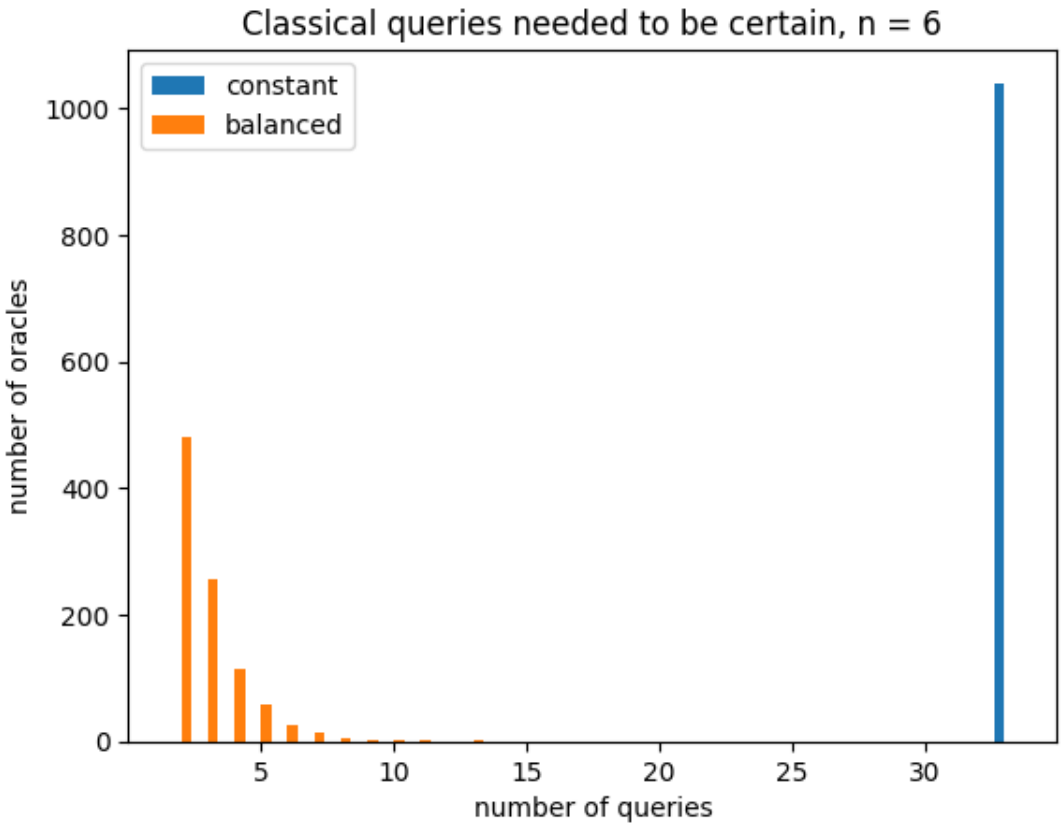
[9]:
```
n = 6
queries = {"constant": [], "balanced": []}
for seed in range(2000):
    f = ClassicalOracle(n, seed)
    queries[classical_solver(f)].append(f.queries)

for kind, q in queries.items():
    print(f"{kind}: min = {min(q)}, max = {max(q)} queries")

plt.hist([queries["constant"], queries["balanced"]], bins=np.arange(1.5, 2**(n␣
 ↪- 1) + 2.5),
         label=["constant", "balanced"])
```

```
plt.xlabel("number of queries")
plt.ylabel("number of oracles")
plt.title(f"Classical queries needed to be certain, n = {n}")
plt.legend()
plt.show()
```

```
constant: min = 33, max = 33 queries
balanced: min = 2, max = 13 queries
```



Q2. For $n = 6$, what is the minimum and the maximum number of queries a classical computer needs to be *certain*? Which kind of function forces the maximum? What is the general formula for n bits?

[10]:
```
# Q2: min = 2 (balanced, the first two answers already differ); max = 2**(n-1)
↪+ 1 = 33.
# A constant function always forces the maximum: only after 2**(n-1) + 1 equal
↪answers (more than
# half of the 2**n inputs) can we exclude "balanced". An unlucky balanced
↪function can also need 33,
```

```
# when the first 2**(n-1) answers are all equal. In general: min = 2, max =
 ↪2**(n-1) + 1 queries.
def query_bounds_N6():
    min_nr_inputs = 2   # first two answers differ -> balanced
    max_nr_inputs = 33  # 2**(6-1) + 1 equal answers needed to prove constant
    return [min_nr_inputs, max_nr_inputs]
```

Optional — being *almost* sure. If we accept a small chance of error, a classical computer can do much better: query k random inputs and answer "constant" if all results are equal. For a balanced function this is wrong with probability about $1/2^{k-1}$ — only about 0.2% for $k = 10$. The advantage of the quantum algorithm in this lab is that it is **100% certain** after a single query.

1.4 Part 2) The Deutsch algorithm

1.4.1 2.1 A quantum oracle for a 1-bit function

A quantum gate must be reversible, but $x \mapsto f(x)$ is not (for f_0 both inputs give 0, so we cannot recover x). The quantum oracle therefore keeps x and adds the answer into an extra qubit y:

$$U_f\,|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

where $\oplus$ is addition modulo 2 (XOR).

We put x on qubit q_0 and y on qubit q_1 — the same layout that the Deutsch-Jozsa oracle uses later (inputs on $q_0 \dots q_{n-1}$, output on q_n). In formulas we write $|x\rangle|y\rangle$; Qiskit writes the same state as $|q_1 q_0\rangle = |y\,x\rangle$.

f	$f(0)$	$f(1)$	circuit for U_f
$f_0(x) = 0$	0	0	do nothing
$f_1(x) = 1$	1	1	X on q_1
$f_2(x) = x$	0	1	CNOT from q_0 to q_1
$f_3(x) = \neg x$	1	0	**Exercise 2**

Exercise 2: Build the oracle for $f_3(x) = \neg x$

Complete case == 3 in deutsch_oracle so that $U_{f_3}|x\rangle|y\rangle = |x\rangle|y \oplus \neg x\rangle$.

Hint: $y \oplus \neg x = (y \oplus x) \oplus 1$.

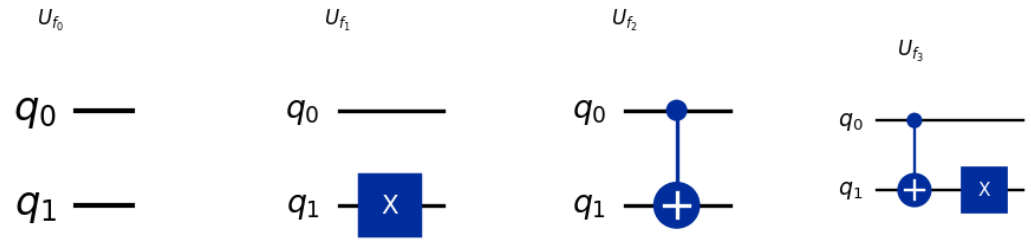
[11]:
```
def deutsch_oracle(case):
    """Quantum oracle U_f for the 1-bit function f_case.
    q0 = x (input), q1 = y (output) -- the same layout as the Deutsch-Jozsa
 ↪oracle."""
    qc = QuantumCircuit(2, name=f"U_f{case}")
    if case == 1:           # f1(x) = 1
        qc.x(1)
    elif case == 2:         # f2(x) = x
        qc.cx(0, 1)
    elif case == 3:         # f3(x) = NOT x
```

```
        ### FILL YOUR CODE IN HERE
        qc.cx(0, 1)         # y XOR x
        qc.x(1)             # ... XOR 1
        ###
    return qc               # case 0: f0(x) = 0 -> do nothing
```

[12]:
```
# Check
check_deutsch_oracle(deutsch_oracle)

fig, axes = plt.subplots(1, 4, figsize=(12, 2.5))
for case, ax in enumerate(axes):
    deutsch_oracle(case).draw(output='mpl', ax=ax)
    ax.set_title(f"$U_{{f_{case}}}$")
plt.show()
```

```
deutsch_oracle is correct for all four functions
```



Using the quantum oracle classically. If we feed in a classical input $|x\rangle|0\rangle$, the oracle just writes $f(x)$ into q_1. Then it is no better than the classical black box: we still need **two runs** to learn $f(0)$ and $f(1)$.

[13]:
```
def query_classically(oracle, x):
    """Feed the classical input |x>|0> into the oracle and measure y, which␣
 ↪gives f(x)."""
    qc = QuantumCircuit(2, 1)
    if x == 1:
        qc.x(0)
    qc.compose(oracle, inplace=True)
    qc.measure(1, 0)
    counts = run(qc)
    return int(max(counts, key=counts.get))

for case in range(4):
    U = deutsch_oracle(case)
    print(f"f{case}: f(0) = {query_classically(U, 0)}, f(1) =␣
 ↪{query_classically(U, 1)}")
```

```
f0: f(0) = 0, f(1) = 0

f1: f(0) = 1, f(1) = 1

f2: f(0) = 0, f(1) = 1
f3: f(0) = 1, f(1) = 0
```

Optional. Print the matrix of U_{f_2}. Every column contains a single 1: the oracle only permutes the basis states, which is why it is reversible. (Qiskit orders the basis as $|q_1 q_0\rangle = 00, 01, 10, 11$.)

[14]:
```
print(Operator(deutsch_oracle(2)).data.real.astype(int))
```

```
[[1 0 0 0]
 [0 0 0 1]
 [0 0 1 0]
 [0 1 0 0]]
```

1.4.2 2.2 Phase kickback

Now prepare the output qubit in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. As shown in the lecture,

$$U_f\,|x\rangle|-\rangle = (-1)^{f(x)}\,|x\rangle|-\rangle$$

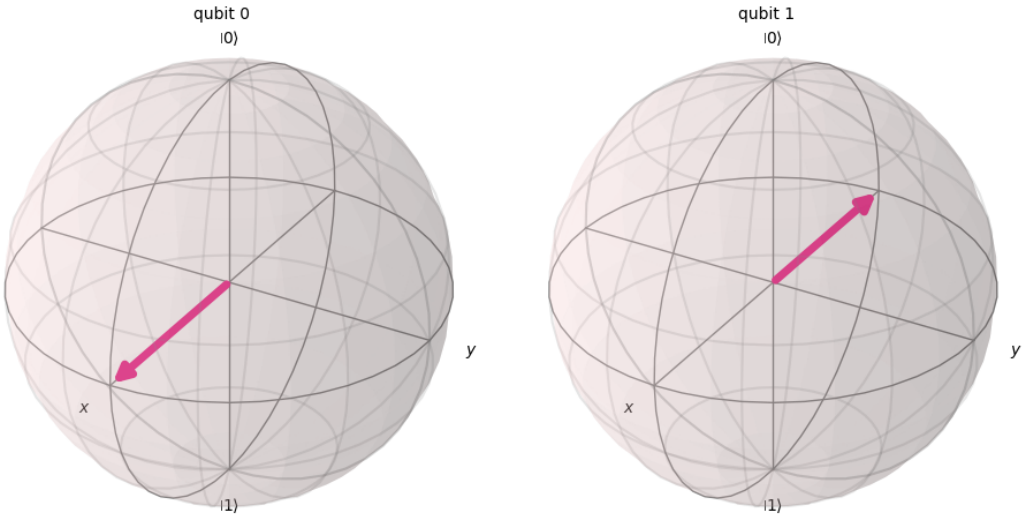
The value of $f(x)$ is "kicked back" as a phase onto the input qubit. If the input is the superposition $|+\rangle$:

$$U_f\,|+\rangle|-\rangle = \frac{1}{\sqrt{2}}\Big((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\Big)|-\rangle = \begin{cases} \pm\,|+\rangle|-\rangle & f \text{ constant} \\ \pm\,|-\rangle|-\rangle & f \text{ balanced} \end{cases}$$

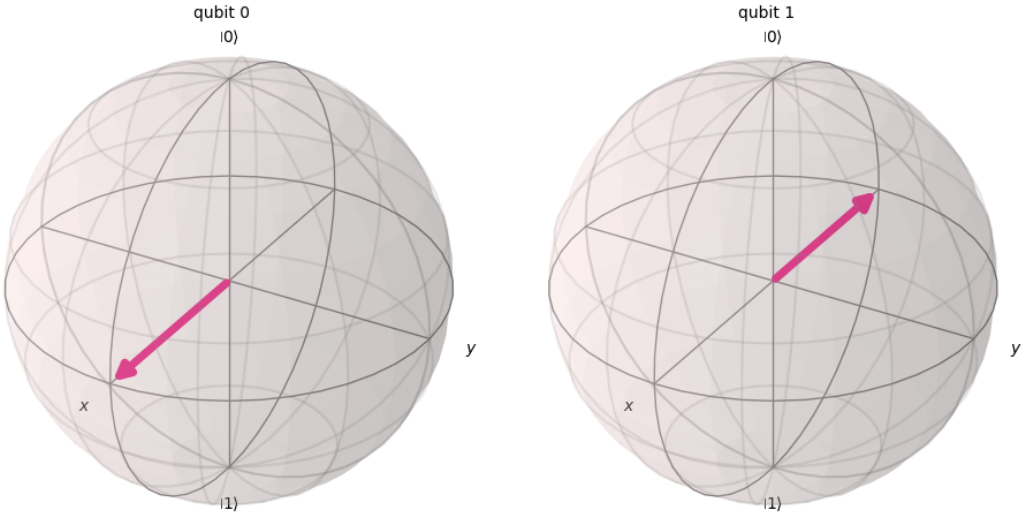
Run the cell to see both qubits after the oracle, for all four functions.

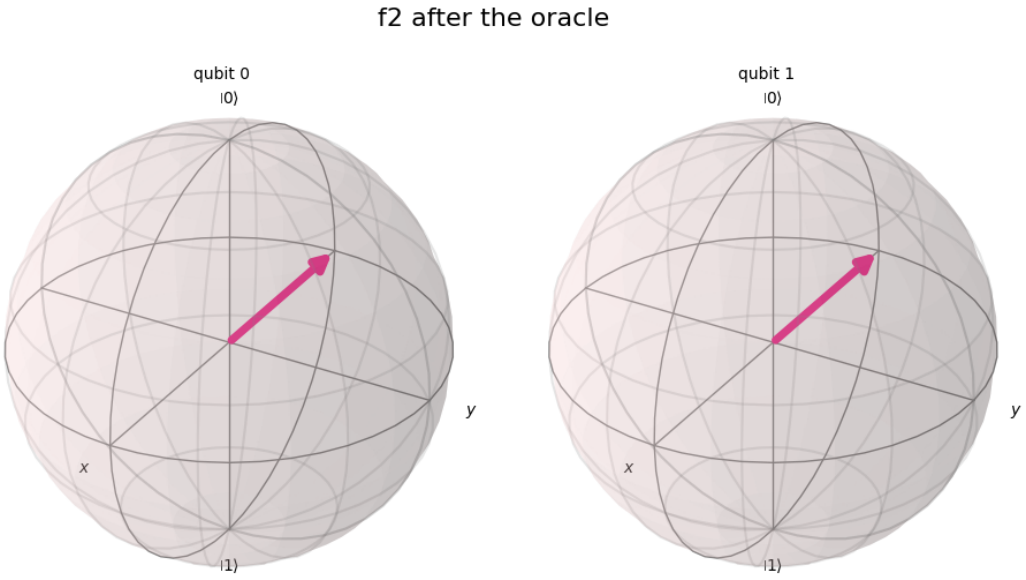
[15]:
```
for case in range(4):
    kickback = QuantumCircuit(2)
    kickback.x(1)
    kickback.h(1)       # y = |->
    kickback.h(0)       # x = |+>
    kickback.compose(deutsch_oracle(case), inplace=True)
    display(plot_bloch_multivector(Statevector(kickback), title=f"f{case} after␣
 ↪the oracle"))
```

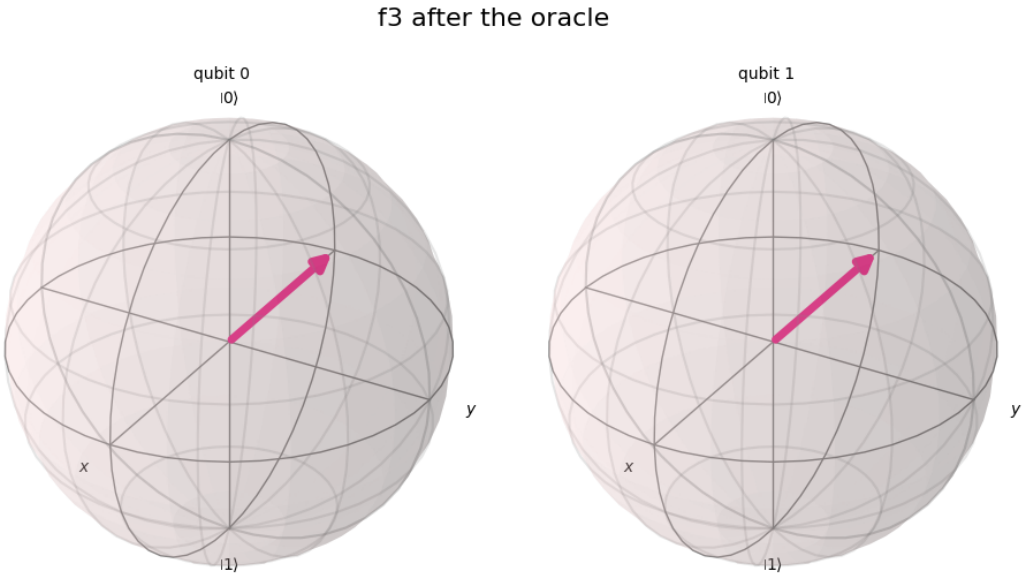
f0 after the oracle



f1 after the oracle







Q3. Where does the Bloch vector of q_0 point for the constant functions, and for the balanced ones? Which single gate turns this difference into a measurement result of 0 or 1? Why do f_0 and f_1 look exactly the same?

Your answer:

- **Constant** (f_0, f_1): the Bloch vector of q_0 points along $+x$, i.e. q_0 is in $|+\rangle$.

- **Balanced** (f_2, f_3): the Bloch vector of q_0 points along $-x$, i.e. q_0 is in $|-\rangle$.
- In all four cases q_1 stays in $|-\rangle$ (pointing along $-x$).

A **Hadamard gate** on q_0 maps $|+\rangle \to |0\rangle$ and $|-\rangle \to |1\rangle$, so measuring q_0 afterwards gives 0 for constant and 1 for balanced.

f_0 and f_1 look exactly the same because $U_{f_1}|+\rangle|-\rangle = -U_{f_0}|+\rangle|-\rangle$: the states differ only by a **global phase** -1 (the phase $(-1)^{f(x)}$ is the same for both inputs). A global phase is unobservable, so the Bloch sphere (and any measurement) cannot distinguish them. The same holds for f_2 and f_3.

1.4.3 2.3 The full Deutsch circuit

Exercise 3: Build the Deutsch circuit

Complete deutsch_circuit(oracle). The steps have the same names as in the Deutsch-Jozsa part later:

1. $|\psi_0\rangle = |-\rangle \otimes |0\rangle$ (Qiskit order $q_1 \otimes q_0$)
2. $|\psi_1\rangle = |-\rangle \otimes |+\rangle$
3. $|\psi_2\rangle = U_f\,|\psi_1\rangle$ — one query (already written for you)
4. Hadamard on q_0, then measure q_0 into classical bit 0

[16]:
```
def deutsch_circuit(oracle):
    qc = QuantumCircuit(2, 1)

    # |psi_0> = |-> (x) |0>
    ### FILL YOUR CODE IN HERE
    qc.x(1)
    qc.h(1)
    ###

    # |psi_1> = |-> (x) |+>
    ### FILL YOUR CODE IN HERE
    qc.h(0)
    ###

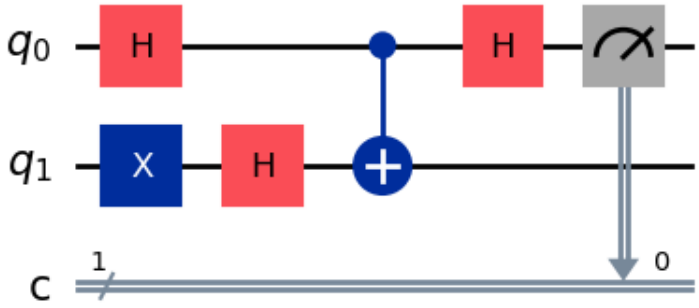
    # |psi_2> : apply the oracle ONCE
    qc.compose(oracle, inplace=True)

    # Hadamard on q0, then measure q0
    ### FILL YOUR CODE IN HERE
    qc.h(0)
    qc.measure(0, 0)
    ###
    return qc
```

[17]:
```
# Check
check_deutsch_circuit(deutsch_circuit)
deutsch_circuit(deutsch_oracle(2)).draw(output='mpl')
```

```
f0 (constant): {'0': 1024}
f1 (constant): {'0': 1024}
f2 (balanced): {'1': 1024}
f3 (balanced): {'1': 1024}
deutsch_circuit is correct: one query tells constant (0) from balanced (1)
```

[17]:



One query, 100% certain: measuring 0 means constant and 1 means balanced. Classically we needed 2 queries.

Optional — mystery oracle. `deutsch_mystery_oracle(seed)` hides one of the four oracles inside a gate. Query it **once** (`shots=1`) and decide whether it is constant or balanced. Try a few seeds.

[18]:
```python
def deutsch_mystery_oracle(seed):
    gate = deutsch_oracle(random.Random(seed).randint(0, 3)).to_gate()
    gate.name = "Mystery"
    return gate

mystery = deutsch_mystery_oracle(seed=2026)    # try other seeds too

# Query the mystery oracle exactly once and decide: constant or balanced?
### FILL YOUR CODE IN HERE
for seed in [2026, 1, 2, 3, 4, 5]:
    counts = run(deutsch_circuit(deutsch_mystery_oracle(seed)), shots=1)
    result = next(iter(counts))
    print(f"seed = {seed}: measured {result} -> {'constant' if result == '0'␣
 ↪else 'balanced'}")
###
```

```
seed = 2026: measured 0 -> constant
seed = 1: measured 0 -> constant
seed = 2: measured 0 -> constant
```

```
seed = 3: measured 0 -> constant
seed = 4: measured 0 -> constant
seed = 5: measured 1 -> balanced
```

1.5 Part 3) The Deutsch-Jozsa problem

Now the hidden Boolean function takes a string of n bits:

$$f(\{x_0, x_1, \ldots, x_{n-1}\}) \rightarrow 0 \text{ or } 1$$

and is again guaranteed to be either constant or balanced. The Deutsch problem from Part 2 is the case $n = 1$.

The Deutsch-Jozsa algorithm was the first example of a quantum algorithm that performs better than the best classical algorithm. It showed that there can be advantages to using a quantum computer as a computational tool for a specific problem.

The function below builds one of many hidden oracles: `problem` selects the oracle and `N` is the number of input bits.

[19]:
```python
def dj_problem_oracle(problem, N, to_gate=True):
    """U_f on N+1 qubits: inputs q_0 ... q_{N-1}, output q_N. Use N >= 2."""
    qc = QuantumCircuit(N+1)
    np.random.seed(problem)
    if np.random.randint(0, 2):
        qc.x(N)

    on_states = np.random.choice(range(2**N), 2**N // 2, replace=False)

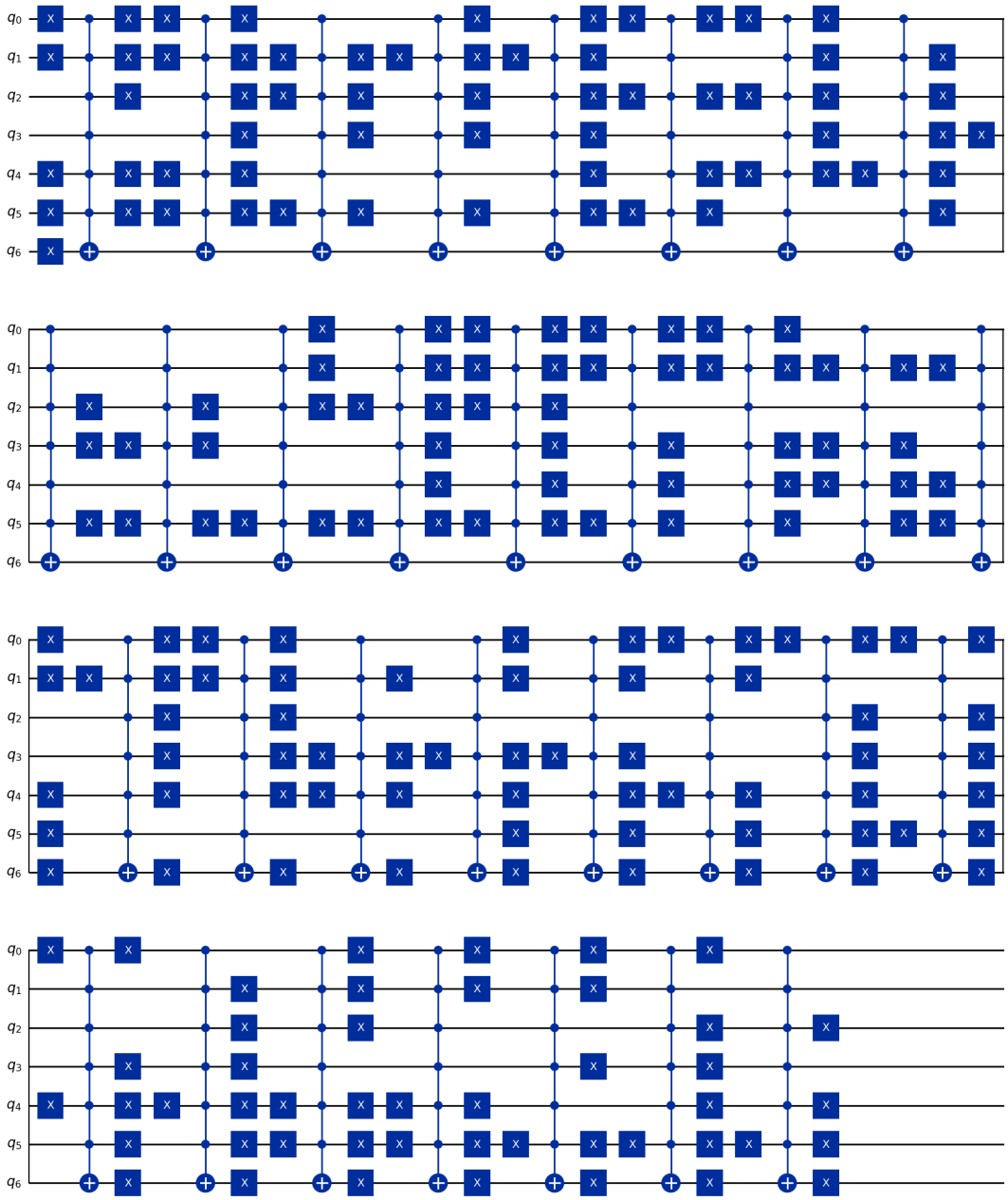
    def add_cx(qc, bit_string):
        for qubit, bit in enumerate(reversed(bit_string)):
            if bit == "1":
                qc.x(qubit)
        return qc

    for state in on_states[:2**N//4]:
        qc = add_cx(qc, f"{state:0b}")
        qc.mcx(list(range(N)), N)
        qc = add_cx(qc, f"{state:0b}")
    if np.random.randint(0, 2):
        for state in on_states[2**N//4:]:
            qc = add_cx(qc, f"{state:0b}")
            qc.mcx(list(range(N)), N)
            qc = add_cx(qc, f"{state:0b}")
            qc.x(N)
    else:
        for state in on_states[2**N//4-1::-1]:
            qc = add_cx(qc, f"{state:0b}")
            qc.mcx(list(range(N)), N)
            qc = add_cx(qc, f"{state:0b}")
```

```
    if to_gate:
        return qc.to_gate()
    else:
        return qc
```

[20]:
```
oraclenr = 1  # determines the oracle
N = 6
oracle = dj_problem_oracle(oraclenr, N)  # gives one out of many oracles
oracle.name = "DJ-Oracle"
dj_problem_oracle(oraclenr, N, False).draw(output='mpl')
```

[20]:



This oracle has $N = 6$ input qubits $q_0 \ldots q_5$; the last qubit q_6 is the output. Just as in Part 2, we can use it classically: prepare one input bit string, apply the oracle, and measure q_6.

Note: here `input_str[i]` sets qubit q_i, so the **leftmost** character is q_0 — the opposite of the order in which Qiskit prints bit strings.

[21]:
```
def dj_classical(oracle, n, input_str):
```

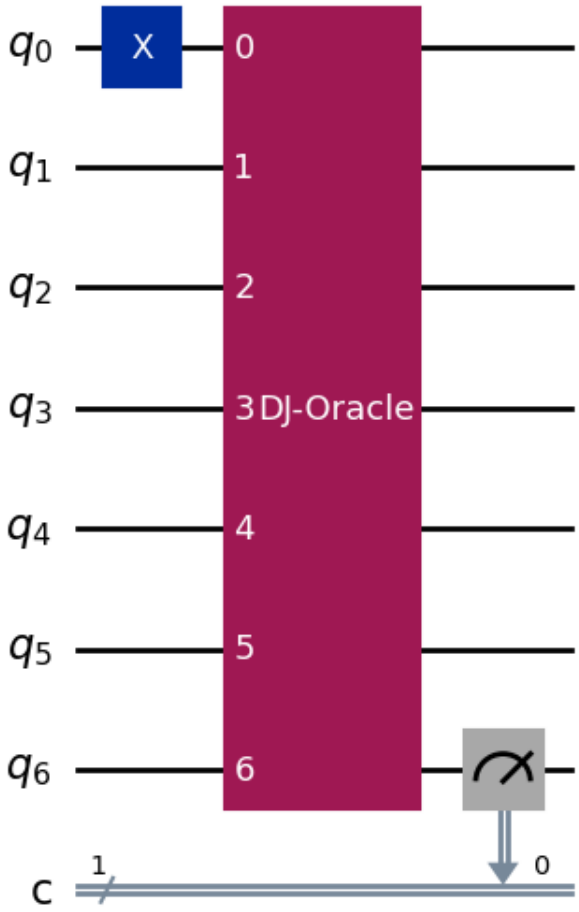
```
    """Query the oracle with ONE classical input string; the measured bit is␣
 ↪f(input_str)."""
    qc = QuantumCircuit(n+1, 1)
    for i in range(n):              # input_str[i] -> qubit q_i
        if input_str[i] == '1':
            qc.x(i)
    qc.append(oracle, range(n+1))
    qc.measure(n, 0)                # measure the output qubit q_n
    return qc

for input_str in ['000000', '100000', '111111', '010010']:
    counts = run(dj_classical(oracle, N, input_str))
    print(f"f({input_str}) = {max(counts, key=counts.get)}")

dj_classical(oracle, N, '100000').draw(output='mpl')
```

```
f(000000) = 0
f(100000) = 1
f(111111) = 1
f(010010) = 1
```

[21]:



Q4. From these four queries, are you *sure* whether oracle 1 is constant or balanced? In the worst case, how many queries could a classical computer need for this oracle (see Q2)?

Your answer:

Yes, in this case we are **sure** that oracle 1 is **balanced**: $f(000000) = 0$ but $f(100000) = 1$. A constant function gives the same value for every input, so two different answers prove "balanced". (This agrees with the quantum result for Problem 1 below.)

We were lucky, though. If all four answers had been equal, we could not have decided. In the worst case (see Q2), a classical computer needs $2^{N-1} + 1 = 2^5 + 1 = 33$ queries for this $N = 6$ oracle.

1.6 Part 4) The Deutsch-Jozsa algorithm

Using a quantum computer, we can decide whether the oracle is constant or balanced with 100% confidence after only **one** call to the oracle $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$. The circuit is the Deutsch circuit

from Part 2.3 with the single input qubit replaced by n input qubits:

	Deutsch (Part 2)	Deutsch-Jozsa
qubits	input q_0, output q_1	inputs $q_0 \dots q_{n-1}$, output q_n
$\|\psi_0\rangle$	$\|-\rangle \otimes \|0\rangle$	$\|-\rangle \otimes \|0\rangle^{\otimes n}$
$\|\psi_1\rangle$	H on q_0	H on each of $q_0 \dots q_{n-1}$
after the oracle	H on q_0, measure q_0	H on each input, measure $q_0 \dots q_{n-1}$
constant if	result is 0	result is all 0

Remember that Qiskit writes states as $|b_n \dots b_1 b_0\rangle$, i.e. the last bit b_0 is the state of qubit q_0. So we want to initialise

$$|\psi_0\rangle = |-\rangle \otimes |0\rangle^{\otimes n}.$$

Exercise 4: Prepare $|\psi_0\rangle$

Complete psi_0(n) so that it builds $|\psi_0\rangle = |-\rangle \otimes |0\rangle^{\otimes n}$: the state $|-\rangle$ on the output qubit q_n and $|0\rangle$ on the input qubits $q_0 \dots q_{n-1}$.

Hint: this is the first step of deutsch_circuit, but the output qubit is now q_n.

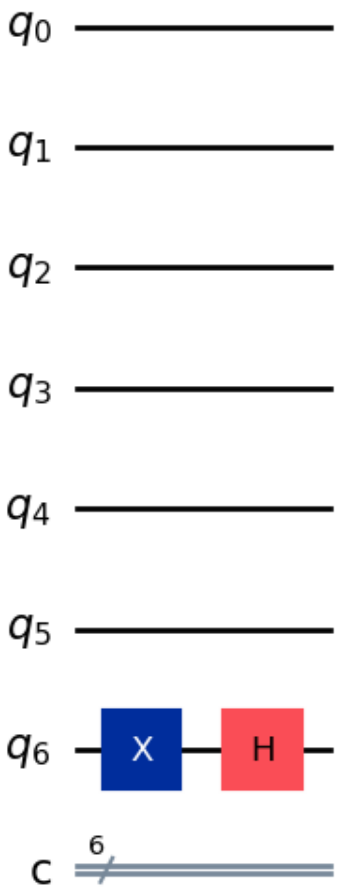
[22]:
```
n = N
def psi_0(n):
    qc = QuantumCircuit(n+1, n)

    # Build the state |-> (x) |0...0>   (|-> on q_n, |0> on q_0 ... q_{n-1})
    ### FILL YOUR CODE IN HERE
    qc.x(n)
    qc.h(n)
    ###
    return qc
```

[23]:
```
# Check
check_psi_0(psi_0, n)
psi_0(n).draw(output='mpl')
```

```
psi_0 is correct
```

[23]:



Next, put the inputs into an equal superposition:

$$|\psi_1\rangle = |-\rangle \otimes \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

Exercise 5: Create the superposition $|\psi_1\rangle$

Complete psi_1(n): start from psi_0(n), then put every input qubit $q_0 \ldots q_{n-1}$ into superposition.

Hint: qc.h() also accepts a list or a range of qubits.

[24]:
```
def psi_1(n):

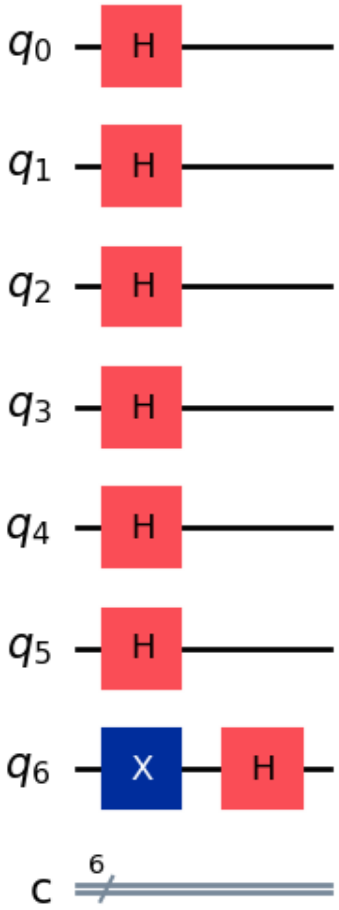
    # obtain the |psi_0> state, then create the superposition state |psi_1>
    ### FILL YOUR CODE IN HERE
    qc = psi_0(n)
```

```
    qc.h(range(n))
    ###
    return qc
```

[25]:
```
# Check
check_psi_1(psi_1, n)
psi_1(n).draw(output='mpl')
```

```
psi_1 is correct
```

[25]:



Apply the oracle **once**. By phase kickback (Part 2.2) this gives

$$|\psi_2\rangle = |-\rangle \otimes \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle$$

Exercise 6: Apply the oracle to get $|\psi_2\rangle$

Complete psi_2(oracle, n): start from psi_1(n) and apply the oracle once. The oracle is a gate that acts on all $n+1$ qubits.

Hint: a gate is added to a circuit with qc.append(gate, qubits).

[26]:
```
def psi_2(oracle, n):

    # circuit to obtain psi_1
    ### FILL YOUR CODE IN HERE
    qc = psi_1(n)
    ###

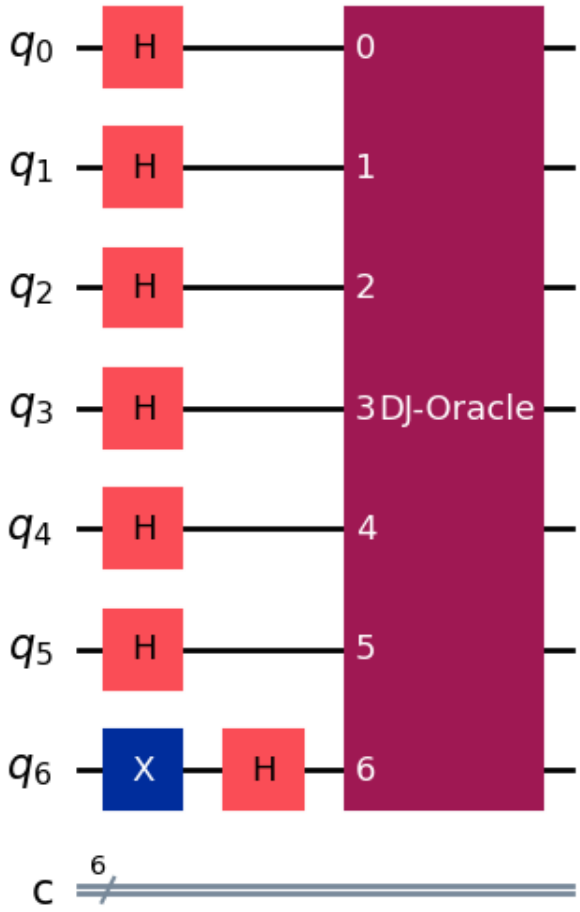
    # append the oracle
    ### FILL YOUR CODE IN HERE
    qc.append(oracle, range(n + 1))
    ###

    return qc
```

[27]:
```
# Check
check_psi_2(psi_2, oracle, n)
psi_2(oracle, n).draw(output='mpl')
```

```
psi_2 is correct
```

[27]:



Finally, we ignore the output qubit q_n, apply a Hadamard to each input qubit and measure the inputs.

Exercise 7: Complete the Deutsch-Jozsa circuit

Complete full_circuit(oracle, n): start from psi_2(oracle, n), apply a Hadamard to every input qubit, then measure every input qubit q_i into classical bit i. The output qubit q_n is not measured.

[28]:
```
def full_circuit(oracle, n):  # uses psi_0, psi_1 and psi_2 from above

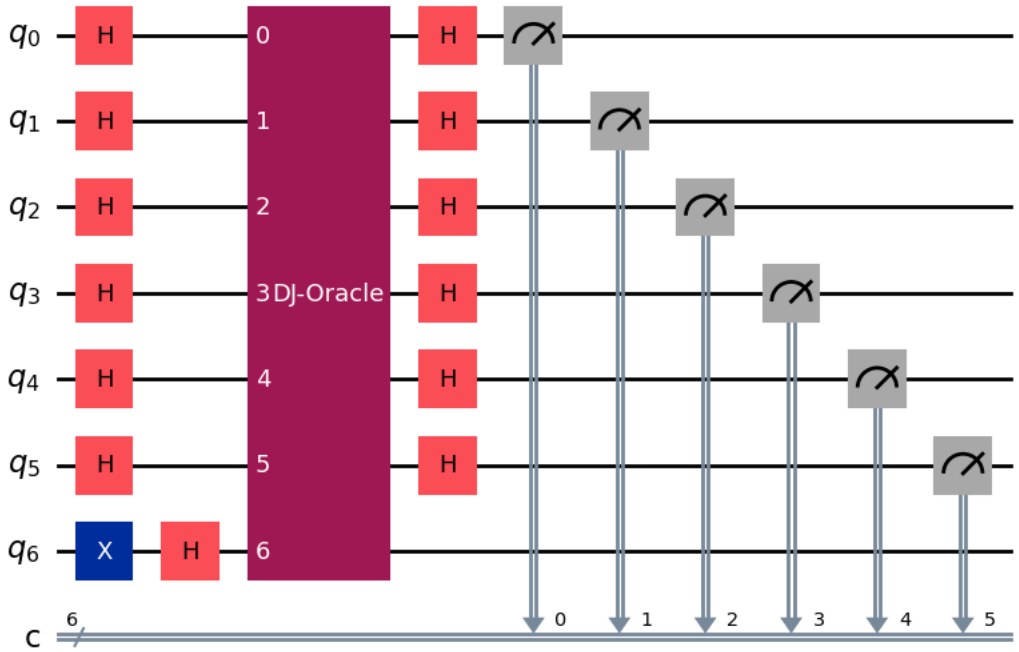
    # get psi_2 and apply n-fold hadamard gate
    ### FILL YOUR CODE IN HERE
    qc = psi_2(oracle, n)
    qc.h(range(n))
    ###
```

```
    # add the measurement by connecting qubits to classical bits
    ### FILL YOUR CODE IN HERE
    qc.measure(range(n), range(n))
    ###
    return qc
```

[29]:
```
# Check
check_full_circuit(full_circuit, n)
dj_circuit = full_circuit(oracle, n)
dj_circuit.draw(output='mpl')
```

```
full_circuit is correct
```

[29]:



For 6 input qubits the circuit should look like this:

As shown in the lecture, the probability of measuring $|0\rangle^{\otimes n}$ is

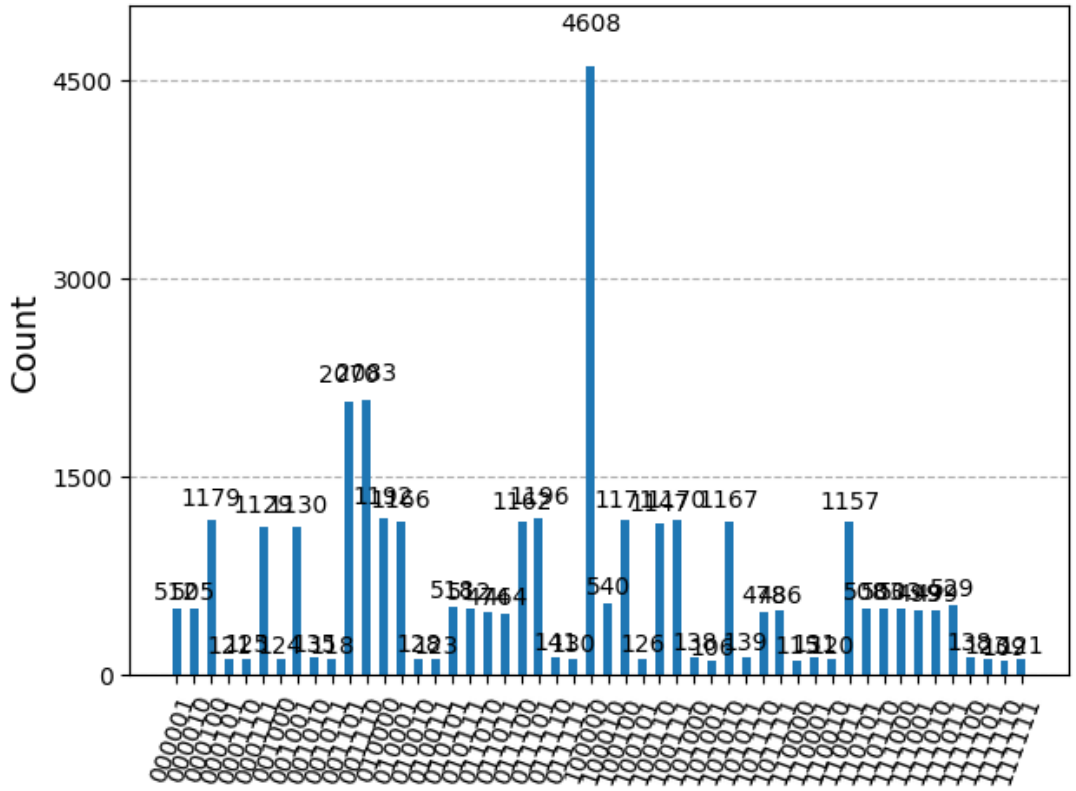
$$\left|\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{f(x)}\right|^2 = \begin{cases} 1 & \text{if } f \text{ is constant} \\ 0 & \text{if } f \text{ is balanced} \end{cases}$$

so **all zeros → constant, anything else → balanced**. (The full derivation is in the appendix at the end.)

Let us run the circuit on the simulator:

[30]:
```
answer = run(dj_circuit, shots=1 << 15)
plot_histogram(answer)
```

[30]:



1.7 Problems 1–10

The cell below runs the Deutsch-Jozsa circuit on oracles 1–10. In principle **one shot (one query)** per oracle is enough; we use 1024 shots only to show that the result is always the same.

Remember that every oracle is guaranteed to be either constant or balanced, so there is no third case to consider. Decide for each oracle which one it is from the measurement result of the first register, and fill in the table.

[31]:
```
for problem in range(1, 11):
    problem_oracle = dj_problem_oracle(problem, N)
    counts = run(full_circuit(problem_oracle, N))
    p_zero = counts.get('0' * N, 0) / 1024
    print(f"Problem {problem:2d}:  P(|{'0' * N}>) = {p_zero:.2f}   most␣
↪frequent outcome: {max(counts, key=counts.get)}")
```

```
Problem  1:  P(|000000>) = 0.00   most frequent outcome: 100000
Problem  2:  P(|000000>) = 1.00   most frequent outcome: 000000
Problem  3:  P(|000000>) = 1.00   most frequent outcome: 000000
```

```
Problem  4:  P(|000000>) = 1.00   most frequent outcome: 000000
Problem  5:  P(|000000>) = 1.00   most frequent outcome: 000000

Problem  6:  P(|000000>) = 0.00   most frequent outcome: 101001
Problem  7:  P(|000000>) = 0.00   most frequent outcome: 000010

Problem  8:  P(|000000>) = 0.00   most frequent outcome: 100110
Problem  9:  P(|000000>) = 0.00   most frequent outcome: 101110
Problem 10:  P(|000000>) = 1.00   most frequent outcome: 000000
```

Problem	$P(\|000000\rangle)$	most frequent outcome	constant or balanced?
1	0.00	`100000`	**balanced**
2	1.00	`000000`	**constant**
3	1.00	`000000`	**constant**
4	1.00	`000000`	**constant**
5	1.00	`000000`	**constant**
6	0.00	`101001`	**balanced**
7	0.00	`000010`	**balanced**
8	0.00	`100110`	**balanced**
9	0.00	`101110`	**balanced**
10	1.00	`000000`	**constant**

1.8 Appendix: why does it work? (reading)

Consider that for any state $|x\rangle = |x_{n-1} \dots x_0\rangle = |x_{n-1}\rangle \otimes \dots \otimes |x_0\rangle$, applying a Hadamard gate to each qubit yields the state:

$$\begin{aligned} H^{\otimes n}|x\rangle &= H|x_{n-1}\rangle \otimes \dots \otimes H|x_0\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x\cdot y}|y\rangle, \end{aligned}$$

where $x \cdot y = x_0y_0 + x_1y_1 + \dots + x_{n-1}y_{n-1}$ is the sum of the bitwise product.

At this point the output qubit may be ignored. Applying a Hadamard gate to each qubit in the input register yields the state:

$$\begin{aligned} |\psi_3\rangle &= \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \left[\sum_{y=0}^{2^n-1} (-1)^{x\cdot y}|y\rangle \right] \\ &= \frac{1}{2^n} \sum_{y=0}^{2^n-1} \left[\sum_{x=0}^{2^n-1} (-1)^{f(x)+x\cdot y} \right] |y\rangle, \end{aligned}$$

Now, consider the amplitude of the basis state $|0\rangle^{\otimes n} = |0 \dots 0\rangle$ in this final state $|\psi_3\rangle$. This amplitude is given by

$$\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)+x\cdot 0} = \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)}.$$

yielding the probability of measuring the state $|0\rangle^{\otimes n}$ as

$$\left| \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \right|^2 = \begin{cases} 1 & \text{if } f \text{ is constant} \\ 0 & \text{if } f \text{ is balanced} \end{cases}$$

In the balanced case the $(-1)^{f(x)}$ terms cancel and the probability of measuring $|0\rangle^{\otimes n}$ is 0. In the constant case $(-1)^{f(x)}$ is the same (+1 or −1) for all x, so the probability is 1.